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Practice Assignment Week 6

1.Create a new directory workflow_lab and inside it prepare at least 5 text files with different extensions. Then write a command that shows only files that start with a or b, and end with .txt or .log.

```
nurarys@nurarys-VirtualBox:~$ mkdir workflow_lab
nurarys@nurarys-VirtualBox:~$ cd workflow_lab
nurarys@nurarys-VirtualBox:~/workflow_lab$ touch a.txt b.log c.md d.cfg notes.csf extra.log
nurarys@nurarys-VirtualBox:~/workflow_lab$ ls -1 [ab]*.{txt,log}
a.txt
b.log
```

2.Start a background process that writes current system time every 5 seconds into a file named time.log, and make sure it continues running even after logout.

```
nurarys@nurarys-VirtualBox:~$ cd ~/workflow_lab
nurarys@nurarys-VirtualBox:~/workflow_lab$ nohup bash -c 'while :; do date +"%F %T"; sleep 5; done >> time.log 2>&1' &
[1] 12589
nurarys@nurarys-VirtualBox:~/workflow_lab$ nohup: ignoring input and appending output to 'nohup.out'
nurarys@nurarys-VirtualBox:~/workflow_lab$ ^[[200~echo $! > time_logger.pid
echo: command not found
nurarys@nurarys-VirtualBox:~/workflow_lab$ echo $! > time_logger.pid
nurarys@nurarys-VirtualBox:~/workflow_lab$ disown
nurarys@nurarys-VirtualBox:~/workflow_lab$ ps -p "$(cat ~/workflow_lab/time_logger.pid)"
    PID TTY          TIME CMD
  12589 pts/0    00:00:00 bash
nurarys@nurarys-VirtualBox:~/workflow_lab$ tail -n3 ~/workflow_lab/time.log
2025-10-14 08:53:22
2025-10-14 08:53:22
2025-10-14 08:53:23
nurarys@nurarys-VirtualBox:~/workflow_lab$
```

3.Check the status of the SSH (or any other active) service on your system, restart it, and confirm that it is enabled on boot.

```
nurarys@nurarys-VirtualBox:~$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: inactive (dead)
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
nurarys@nurarys-VirtualBox:~$ sudo systemctl restart ssh
nurarys@nurarys-VirtualBox:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
nurarys@nurarys-VirtualBox:~$ systemctl is-enabled ssh
enabled
```

4.Write one command that lists all running processes owned by your user, sorted by CPU usage, and redirect this output into process_report.txt.

```
nurarys@nurarys-VirtualBox:~$ ps -u "$USER" -o pid,pcpu,pmem,stat,etime,comm --sort=-pcpu > ~/workflow_lab/process_report.txt
nurarys@nurarys-VirtualBox:~$ head ~/workflow_lab/process_report.txt
    PID %CPU %MEM STAT     ELAPSED COMMAND
  14700  100  0.1 R+         00:00 ps
   3250  2.3 11.7 Ssl        41:39 gnome-shell
  12418  0.4  1.7 Ssl        13:28 gnome-terminal-
   3018  0.1  0.4 Ssl        41:40 pipewire
   3433  0.1  0.3 Ssl        41:37 ibus-daemon
   3613  0.0  0.8 Sl         41:37 ibus-extension-
   3972  0.0  1.8 Sl         41:34 gjs
   3024  0.0  0.5 Ssl        41:40 wireplumber
   3026  0.0  0.4 Ssl        41:40 pipewire-pulse
```

5.Create a scheduled task that automatically appends the current date and username to a file named daily_report.txt every minute. Wait 3–4 minutes and verify that it works.

```
nurarys@nurarys-VirtualBox:~$ ( crontab -l 2>/dev/null; echo '* * * * * echo "$(date "+\%F \%T") $USER" >> ~/workflow_lab/
daily_report.txt' ) | crontab -
nurarys@nurarys-VirtualBox:~$ tail -n5 ~/workflow_lab/daily_report.txt
2025-10-14 09:07:01
2025-10-14 09:08:01
2025-10-14 09:09:02
2025-10-14 09:10:01
2025-10-14 09:11:01
```

6.Send a custom signal to a background process to stop it gracefully, then verify that it is no longer running.

```
nurarys@nurarys-VirtualBox:~$ kill -SIGTERM "$(cat ~/workflow_lab/time_logger.pid)" 2>/dev/null
cat: /home/nurarys/workflow_lab/time_logger.pid: No such file or directory
nurarys@nurarys-VirtualBox:~$ sleep 1
nurarys@nurarys-VirtualBox:~$ ps -p "$(cat ~/workflow_lab/time_logger.pid)" || echo "time logger stopped"
cat: /home/nurarys/workflow_lab/time_logger.pid: No such file or directory
error: list of process IDs must follow -p

Usage:
 ps [options]

 Try 'ps --help <simple|list|output|threads|misc|all>'
  or 'ps --help <s|l|o|t|m|a>'
 for additional help text.

For more details see ps(1).
time logger stopped
```

7.Find and display all processes that contain the word “service” in their name, count them, and append the count to system_summary.txt.

```
nurarys@nurarys-VirtualBox:~$ ps -eo pid,cmd  | grep -i '[s]ervice' | tee /dev/tty | wc -l | awk '{print "service_processes=" $1}'
 >> ~/workflow_lab/system_summary.txt
   1893 /usr/sbin/VBoxService --pidfile /var/run/vboxadd-service.sh
   3209 /usr/libexec/gnome-session-binary --systemd-service --session=ubuntu
   3673 /usr/libexec/goa-identity-service
   3796 /usr/libexec/dconf-service
  16212 awk {print "service_processes=" $1}
nurarys@nurarys-VirtualBox:~$ tail -n1 ~/workflow_lab/system_summary.txt
service_processes=5
```

8.Create a custom service called hello.service that runs a script printing “Hello from Linux Service” every 30 seconds. Enable and start it, then verify that it is active.

Create script

```
nurarys@nurarys-VirtualBox:~$ sudo nano /usr/local/bin/hello.sh
nurarys@nurarys-VirtualBox: ~
  GNU nano 7.2                                  /usr/local/bin/hello.sh *
#!/bin/bash
while true; do
  echo "Hello from Linux Service" >> /tmp/hello.log
  sleep 30
done
```

Create service

```
nurarys@nurarys-VirtualBox:~$ sudo chmod +x /usr/local/bin/hello.sh
nurarys@nurarys-VirtualBox:~$ sudo nano /etc/systemd/system/hello.service
```

```
nurarys@nurarys-VirtualBox: ~
  GNU nano 7.2                                  /etc/systemd/system/hello.service *
[Unit]
Description=Hello Service

[Service]
ExecStart=/usr/local/bin/hello.sh
Restart=always

[Install]
WantedBy=multi-user.target
```

Start service

```
nurarys@nurarys-VirtualBox:~$ sudo systemctl daemon-reload
nurarys@nurarys-VirtualBox:~$ sudo systemctl enable hello.service
Created symlink /etc/systemd/system/multi-user.target.wants/hello.service → /etc/systemd/system/hello.service.
nurarys@nurarys-VirtualBox:~$ sudo systemctl start hello.service
```

Test

```
nurarys@nurarys-VirtualBox:~$ systemctl status hello.service
● hello.service - Hello Service
     Loaded: loaded (/etc/systemd/system/hello.service; enabled; preset: enabled)
     Active: active (running) since Tue 2025-10-14 09:56:48 +05; 2min 47s ago
   Main PID: 17856 (hello.sh)
      Tasks: 2 (limit: 4178)
     Memory: 568.0K (peak: 1.1M)
        CPU: 33ms
     CGroup: /system.slice/hello.service
             ├─17856 /bin/bash /usr/local/bin/hello.sh
             └─17985 sleep 30

Oct 14 09:56:48 nurarys-VirtualBox systemd[1]: Started hello.service - Hello Service.
nurarys@nurarys-VirtualBox:~$ tail -f /tmp/hello.log
Hello from Linux Service
Hello from Linux Service
Hello from Linux Service
Hello from Linux Service
Hello from Linux Service
Hello from Linux Service
Hello from Linux Service
```

9.Write one command that shows all active services and filters only those with “failed” status, saving the result into a file failed_services.log.

```
nurarys@nurarys-VirtualBox:~$ systemctl --failed --type=service --no-pager | tee ~/workflow_lab/failed_services.log
  UNIT            LOAD   ACTIVE SUB    DESCRIPTION
● vboxadd.service loaded failed failed vboxadd.service

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

1 loaded units listed.
```

10. Create a cron job that automatically creates a backup of the /home directory every Sunday at midnight, storing it in /var/backups with a timestamp in the filename.

```
nurarys@nurarys-VirtualBox:~$ ( sudo crontab -l 2>/dev/null; echo '0 0 * * 0 tar -czf /var/backups/home-$(date +\%\F-\%H\%M).tar.gz
 /home' ) | sudo crontab -
[sudo] password for nurarys:
nurarys@nurarys-VirtualBox:~$ sudo crontab -l
0 0 * * 0 tar -czf /var/backups/home-$(date +\%\F-\%H\%M).tar.gz /home
```